

Assignment 2

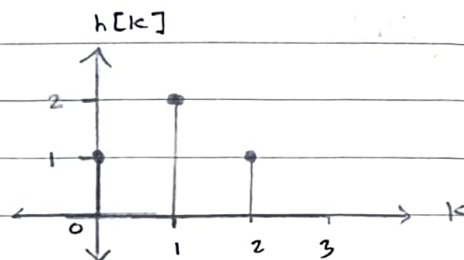
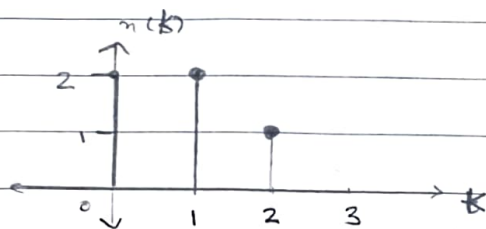
Q > Find $y[n]$, given $x[k] = 2, 2, 1$ $h[k] = 1, 2, 1$

General Formula:

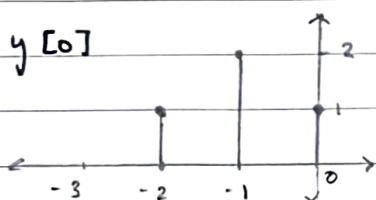
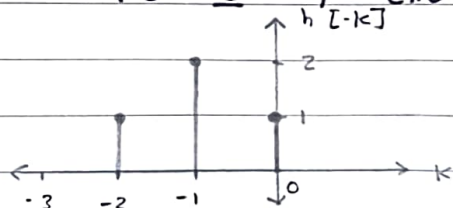
$$y[n] = x[n] * h[n] \rightarrow \text{impulse}$$

input convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

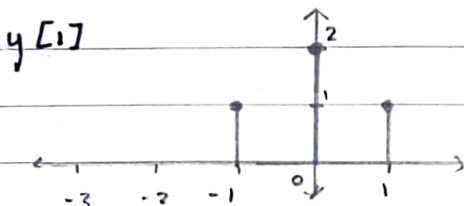


Start point = $0 + 0 = 0$, end point = $2 + 2 = 4$



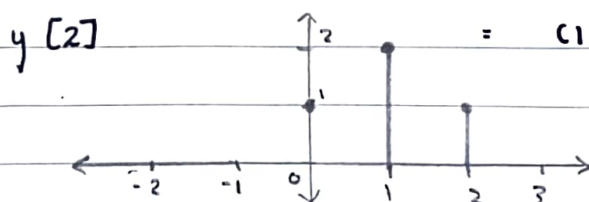
$$y[0] = ((1) \cdot 0) + ((2) \cdot 0) + ((1) \times 2)$$

$$= 2$$



$$y[1] = (0) + (2 \times 2) + (1 \times 2)$$

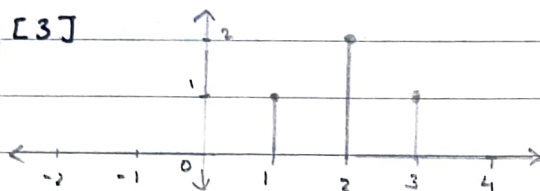
$$= 6$$



$$y[2] = (1 \times 2) + (2 \times 2) + (1 \times 1)$$

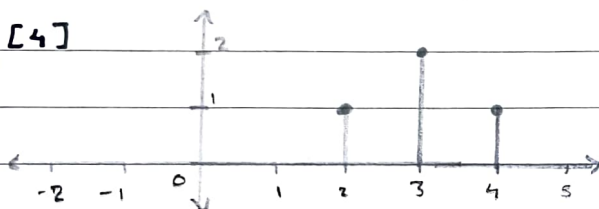
$$= 7$$

$y[3]$

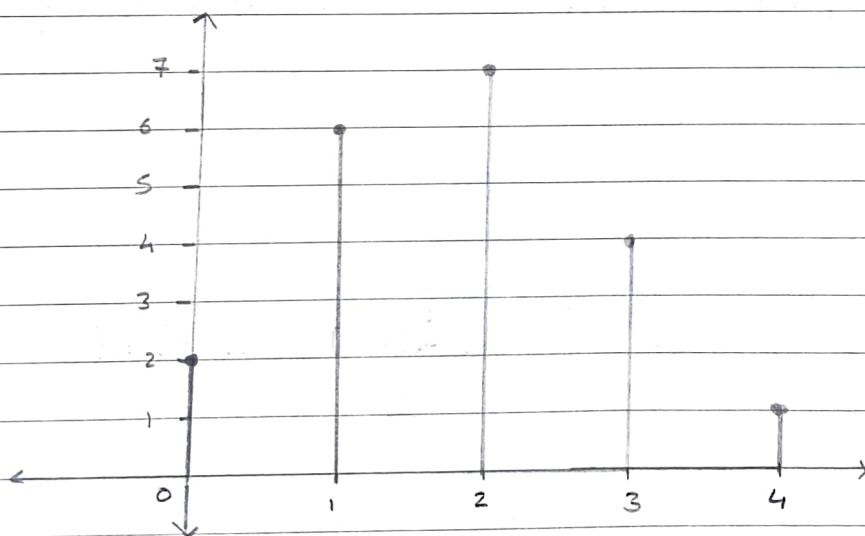


$$\begin{aligned} &= (1 \times 2) + (2 \times 1) + (1 \times 0) \\ &= 4 \end{aligned}$$

$y[4]$



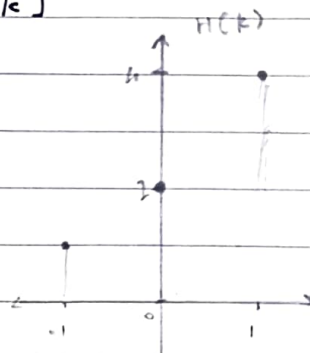
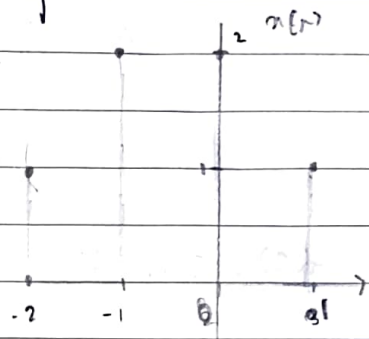
$$\begin{aligned} &= (1 \times 1) + 0 + 6 \\ &= 1 \end{aligned}$$



Find discrete convolution find $y[n]$, given $n[k] = 1, 2, 2, 1$
 $h[k] = 1, 2, 4$

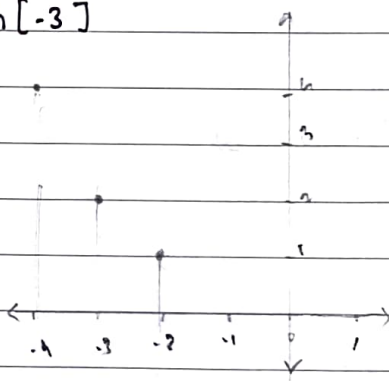
$$y[n] = n[k] * h[k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} n[k] \cdot h[n-k]$$



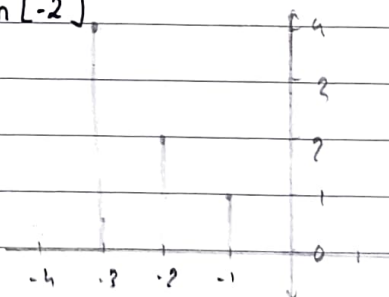
(start point / end point) : start point + start point

$n[-3]$



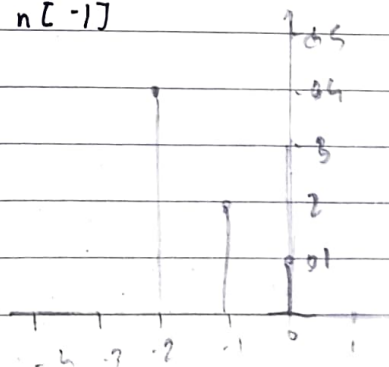
$$= 1 * 1 = \underline{1}$$

$n[-2]$



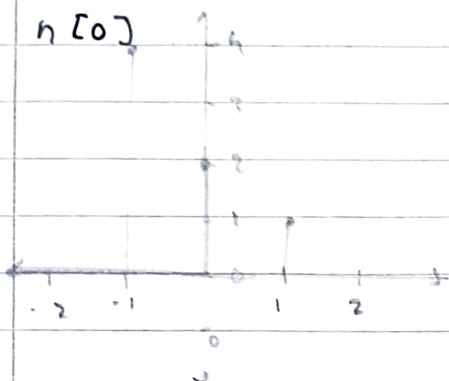
$$= (2 * 1) + (1 * 2) = \underline{4}$$

$n[-1]$



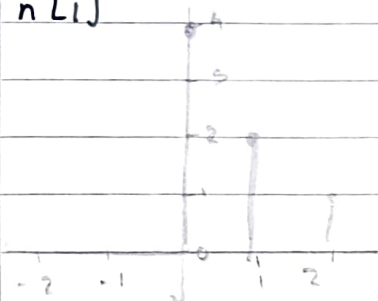
$$= (4 * 1) + (2 * 2) + (1 * 2) = \underline{10}$$

$n[0]$



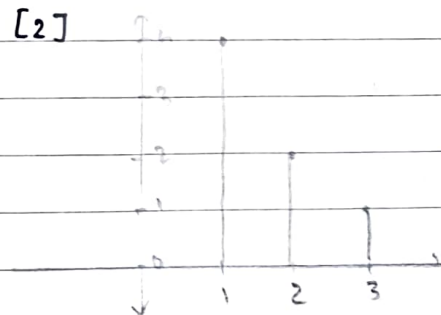
$$= (4 \times 2) + (2 \times 2) + (1 \times 1) = \underline{\underline{13}}$$

$n[1]$



$$= (4 \times 2) + (2 \times 1) = 10$$

$n[2]$



$$= (4 \times 1) = \underline{\underline{4}}$$

unit signal determines lower bound

upper bound

1) Find $x(t) = e^{-t} \cdot u(t)$, $h(t) = e^{-2t} \cdot u(t)$

$$y = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(k) \cdot h(t-k) dk$$

Replace t by k

$$x(k) = e^{-k} \cdot u(k)$$

Replace t by $t-k$

$$h(t-k) = e^{-2(t-k)} \cdot u(t-k)$$

* note using $u(t)$ unit signal it reduces range as negative side is cut.

$$\int_{-\infty}^{\infty} e^{-k} \cdot u(k) \cdot e^{-2t+2k} \cdot u(t-k) dk$$

* note quote equation in brackets to 0
 $k=0$, $t-k=0 \therefore t=k$

* now we eliminat terms having brackets.

$$= \int_0^t e^{-k} \cdot e^{-2t+2k} \cdot dk$$

$$= \int_0^t e^k \cdot e^{-2t} dk$$

$$= e^{-2t} [e^k]_0^t$$

$$= e^{-2t} [e^t - 1]$$

$$= e^{-2t} [e^t - 1]$$

Find $y(t)$ $x(t) = e^{-4t} \cdot u(t)$ $h(t) = u(t-4)$

$$y(t) = x(t) \cdot h(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(t) \cdot h(t-k) dk$$

Replace t by k

$$x(k) = e^{-4k} \cdot u(k)$$

Replace t by $t-k$

$$h(t-k) = u(t-k-4)$$

$$y(t) = \int_{-\infty}^{\infty} e^{-4k} \cdot u(k) \cdot u(t-k-4) dk$$

$$k=0, t-k-4=0 \quad \text{the limits} \quad t-k-4=0$$

$$= \int_0^{t-4} e^{-4k} dk$$

$$k = t-4$$

$$= \left[\frac{e^{-4k}}{-4} \right]_0^{t-4}$$

$$= \left(e^{-4(t-4)} - 1 \right) \cdot \frac{1}{-4}$$

$$= -\frac{1}{4} (e^{-4t+16} - 1)$$

$$= \frac{1}{4} - \frac{e^{-4t+16}}{4}$$

Find $y(t)$

$$x(t) = e^t$$

$$h(t) = (3e^{-2t} - 1)u(t)$$

$$x(k) = e^k$$

$$h(k) = (3e^{-2(t-k)} - 1)u(t-k)$$

$$y(t) = \int_{-\infty}^{\infty} e^k \underbrace{h(k)}_{k=0, t=1} (3e^{-2t+2k} - 1) u(t-k) dk$$

$$\int_{-\infty}^t e^k (3e^{-2t+2k} - 1) dk$$

replace t by k

$$x(k) = e^k$$

$$= \int_{-\infty}^t 3e^{2k-t} - e^k dk$$

$$= -1 \left[3e^{2k-t} \right]_{-\infty}^t - \left[e^k \right]_{-\infty}^t$$

$$= -1 \left[3e^t - 3e^{-t} \right] - \left[e^t - 1 \right]$$

$$\int_{-\infty}^t 3e^{2k} e^{-t} dk$$

$$\int_{-\infty}^t e^k (3e^{-2(t-k)} - 1) u(t-k) dk$$

$$= \int_{-\infty}^t 3e^{-2t+2k} e^k dk - \int_{-\infty}^t e^k dk$$

$$= 3e^{-2t} \left[\frac{e^{3k}}{3} \right]_{-\infty}^t - \left[e^k \right]_{-\infty}^t$$

$$= e^{-2t} \left[e^{3t} - 1 \right] - \left[e^t - 1 \right]$$

$$= e^t - e^{-t}$$

$$= 0$$

example of
example

Explain with an example of convolution

Convolution is a mathematical operation on two functions (f and g) that produces a third function expressing how the shape of one is modified by the other.

Example

- > In digital imaging an image is a 2D grid of kernel that calculates the average of a pixel and its eight neighbours. This softens images and removes noise generally referred to as blurring/smoothing

$$k = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$O = \frac{1}{9} \sum_{i=-1}^1 \sum_{j=-1}^1 I(x+i, y+j)$$

- > Edge Detection (Sharpening): kernels like the Sobel operator or Robert's Cross operator highlight areas with high contrast, enabling computers to find the boundaries of the object.

Horizontal Gradient (G_x):

$$k_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Vertical Gradient (G_y):

$$k_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

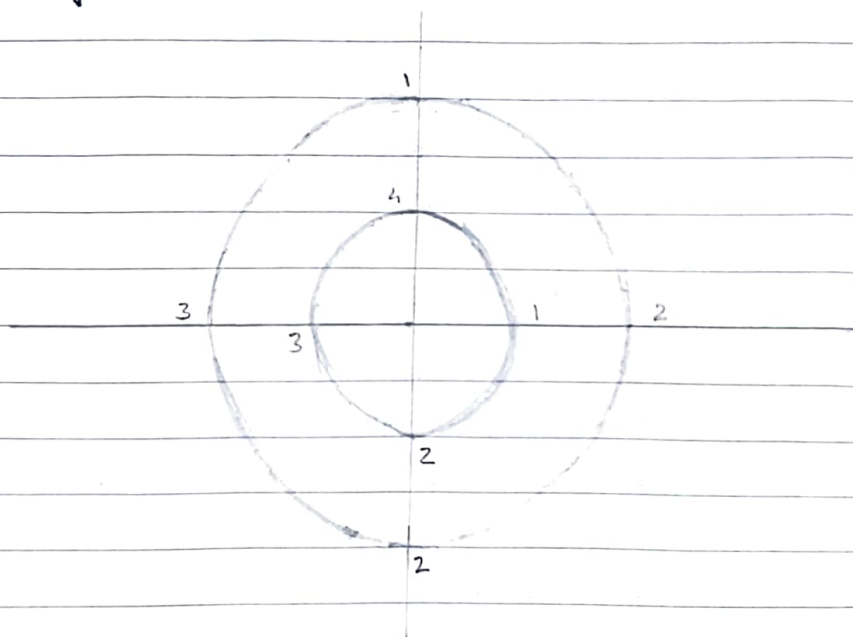
$$\text{Equation} = (G) = \sqrt{G_x^2 + G_y^2}$$

Circular Convolution

Q1> $x[n] = 1, 2, 3, 5, 1, 2, 3, 5$ $L_1 = 4$
 $h[n] = 4, 5, 6, 7, 4, 5, 6, 7$ $L_2 = 4$
 Find $y[n]$

Q2> $x[n] = 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4$ $L_1 = 4$
 $h[n] = 2, 1, 3, 2, 2, 1, 3, 2$ $L_2 = 4$
 Find $y[n]$

$$y[n] = x[n] \cdot h[n]$$

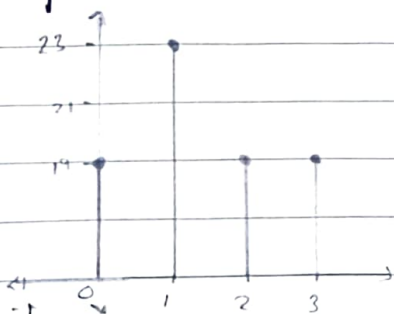


$$y[0] = 1 \times 2 + 2 \times 2 + 3 \times 3 + 4 \times 1 = 2 + 4 + 9 + 4 = 19$$

$$y[1] = 1 \times 1 + 2 \times 2 + 3 \times 2 + 4 \times 3 = 1 + 4 + 6 + 12 = 23$$

$$y[2] = 1 \times 3 + 2 \times 1 + 3 \times 2 + 4 \times 2 = 3 + 2 + 6 + 8 = 19$$

$$y[3] = 1 \times 2 + 2 \times 3 + 3 \times 1 + 4 \times 2 = 2 + 6 + 3 + 8 = 19$$



$$\begin{bmatrix} 2 & 2 & 3 & 1 \\ 1 & 2 & 2 & 3 \\ 3 & 1 & 2 & 2 \\ 2 & 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 19 \\ 23 \\ 19 \\ 19 \end{bmatrix}$$

Q7 $x[n] = 1, 2, 3, 1, 2, 3$

$L_1 = 3$

$h[n] = 2, 1, 2, 1$

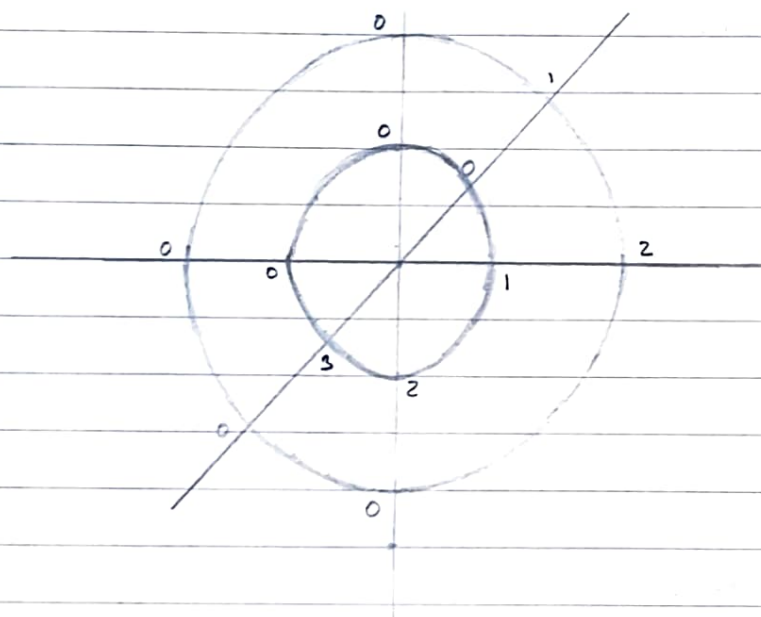
$L_2 = 2$

Find $y[n]$

$LCM(L_1, L_2) = 6$

$x[n] = 1, 2, 3, 0, 0, 0$

$h[n] = 2, 1, 0, 0, 0, 0$



$y[0] = 1 \times 2 + 2 \times 0 + 3 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 1 = 2 + 0 = 2$

$y[1] = 1 \times 1 + 2 \times 2 + 3 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0 = 2 + 4 = 6$

$y[2] = 1 \times 0 + 2 \times 1 + 3 \times 2 + 0 \times 0 + 0 \times 0 + 0 \times 0 = 2 + 6 = 8$

$y[3] = 1 \times 0 + 2 \times 0 + 3 \times 1 + 0 \times 0^2 + 0 \times 0 + 0 \times 0 = 3 = 3$

$y[4] = 1 \times 0 + 2 \times 0 + 3 \times 0 + 0 \times 1 + 0 \times 2 + 0 \times 0 = 0 = 0$

$y[5] = 1 \times 0 + 2 \times 0 + 3 \times 0 + 0 \times 0 + 0 \times 1 + 0 \times 2 = 0 = 0$

